# Structured Noise Impairs Circular Working Memory in a Frozen RNN

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*MSc Dissertation*

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## Abstract

Five independently trained Yang-style fixation-gated RNNs performed continuous circular delayed response. Equal-energy hidden-state perturbations were compared: independent Gaussian noise, temporally correlated AR(1) noise, and AR(1) noise aligned with recurrent topology. These are psilocybin-informed modelling interventions, not literal pharmacology. At matched RMS, impairment followed independent < temporal < topology-correlated structure, grew with delay, accumulated during maintenance, and coincided with collapse of the circular memory geometry. Delay-phase topology noise was far more damaging than cue- or response-phase noise.

## I. QUESTION

Does the *structure* of equal-energy hidden-state noise change how badly a frozen working-memory RNN loses a remembered angle? The comparison was motivated by two literatures. Working-memory models show that stochastic fluctuations can drive drift and that robustness depends on the network's dynamical organization [1, 2]. Psychedelic studies motivate increased signal diversity, temporally structured and context-sensitive activity, and network-topology-dependent changes rather than a simple equation of psychedelics with unstructured noise [3–7]. The three conditions therefore form a controlled hierarchy—independent, temporally persistent, and temporally plus population correlated—without claiming to simulate receptor pharmacology.

## II. WHAT THE THREE NOISES MEAN

Noise $\boldsymbol{\varepsilon}_t$ is added to the leaky recurrent update during selected task phases, then rescaled so every condition has the same realised root-mean-square (RMS) amplitude. RMS is therefore matched energy, not a drug dose.

*a.* *Independent Gaussian.* Each time, trial, and hidden unit is kicked independently:

$$\tilde{\boldsymbol{\varepsilon}}_t \sim \mathcal{N}(\mathbf{0}, I). \tag{1}$$

There is no designed persistence across time and no designed coupling across units. This is the reference stochastic-robustness condition: independent fluctuations are a standard way to test whether a recurrent memory representation diffuses or leaves its task-relevant manifold [1, 2]. It also provides the necessary unstructured control for asking whether adding temporal or population structure changes the effect at the same RMS. It is not presented as a model of psilocybin.

*b.* *Temporal AR(1).* Each unit follows a sticky autoregressive process with $\rho = 0.9$:

$$\mathbf{z}_t = \rho \mathbf{z}_{t-1} + \sqrt{1-\rho^2}\,\boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, I). \tag{2}$$

The expected lag-$k$ autocorrelation is $\rho^k$. Units remain independent; only time is correlated. Intuition: a gust that keeps blowing the same way for a while. This condition isolates *temporal persistence*: unlike independent kicks, a displacement continues in a similar direction over several recurrent updates and can therefore accumulate during memory maintenance. The motivation is that psychedelic-related variability has measurable temporal organization and diversity [4, 6]; AR(1) is a deliberately simple test process, not a process fitted to those data. The choice $\rho = 0.9$ is a project parameter.

*c.* *Topology-correlated AR(1).* The same AR(1) process is then mixed across units using recurrent-weight similarity:

$$G = \frac{W_{\text{rec}} W_{\text{rec}}^\top}{\text{mean}[\text{diag}(W_{\text{rec}} W_{\text{rec}}^\top)]}, \tag{3}$$

$$\Sigma = \frac{0.25I + 0.75G}{\text{mean}[\text{diag}(0.25I + 0.75G)]}, \tag{4}$$

$$\boldsymbol{\varepsilon}_t = L\mathbf{z}_t, \qquad LL^\top = \Sigma. \tag{5}$$

Units with similar incoming recurrent weight profiles tend to be pushed together. This condition tests whether variability aligned with the trained recurrent architecture is more consequential than equally strong unit-wise variability. The motivation is Herzog et al.'s whole-brain result that the spatial pattern of psychedelic-related entropy change depended strongly on structural-network topology, not receptor density alone [5], together with computational evidence that learned network structure can distinguish organized variability from additive

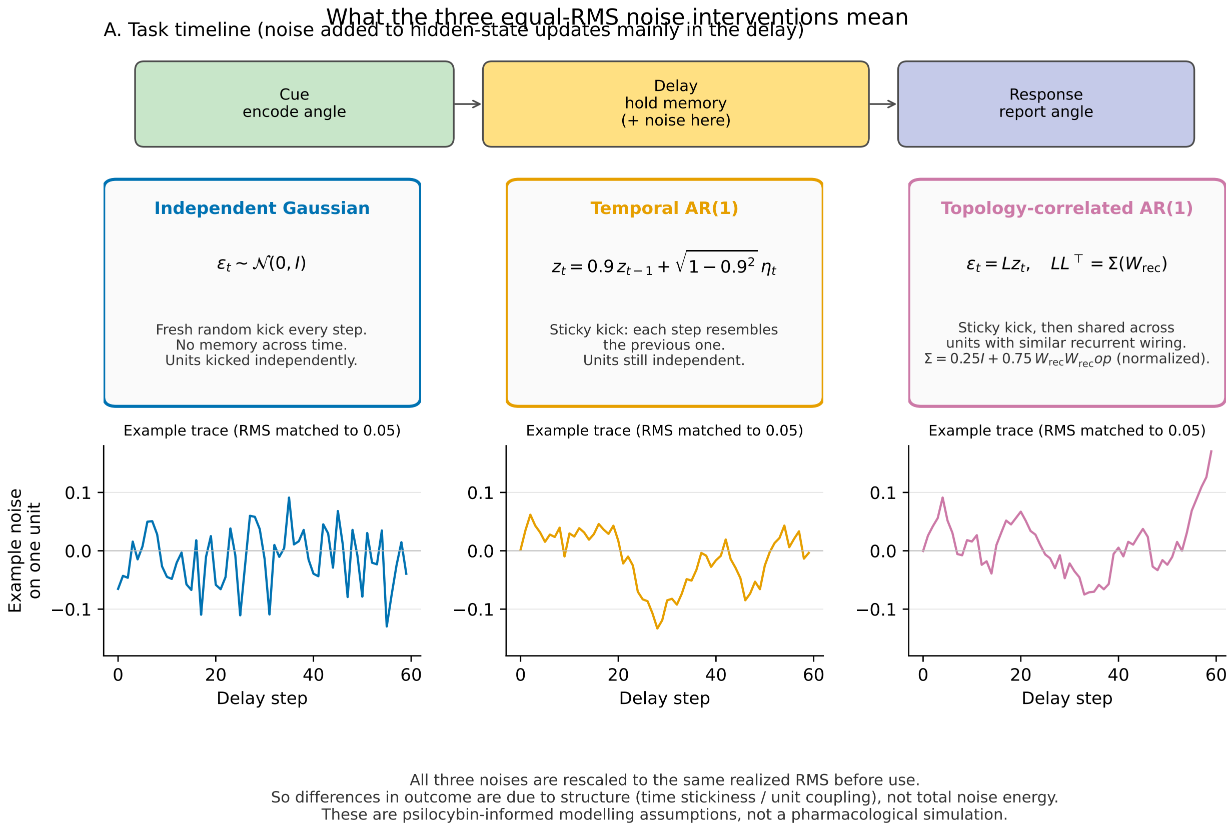


FIG. 1. **Noise definitions.** (A) Cue–delay–response task; main analyses add noise during delay. Below: equations, plain-language meaning, and matched-RMS example traces for the three interventions.

noise [7]. The covariance $0.25I + 0.75G$ is our project operationalisation of that question: it is neither taken from those papers nor a receptor or connectome map.

Fig. 1 summarises the task timeline and these definitions. Fig. 2 checks that RMS matched while temporal stickiness and unit coupling differed as designed. Correlation heatmaps were replaced by a direct summary of mean absolute unit–unit correlation, which is the quantity a marker needs.

## III. MAIN BEHAVIOURAL RESULT

At RMS 0.05, response error increased with structure and delay (Fig. 3A; Table I). At delay 80 the five-seed means were $6.49 \pm 1.93°$ (clean), $7.47 \pm 1.68°$ (independent), $16.33 \pm 2.71°$

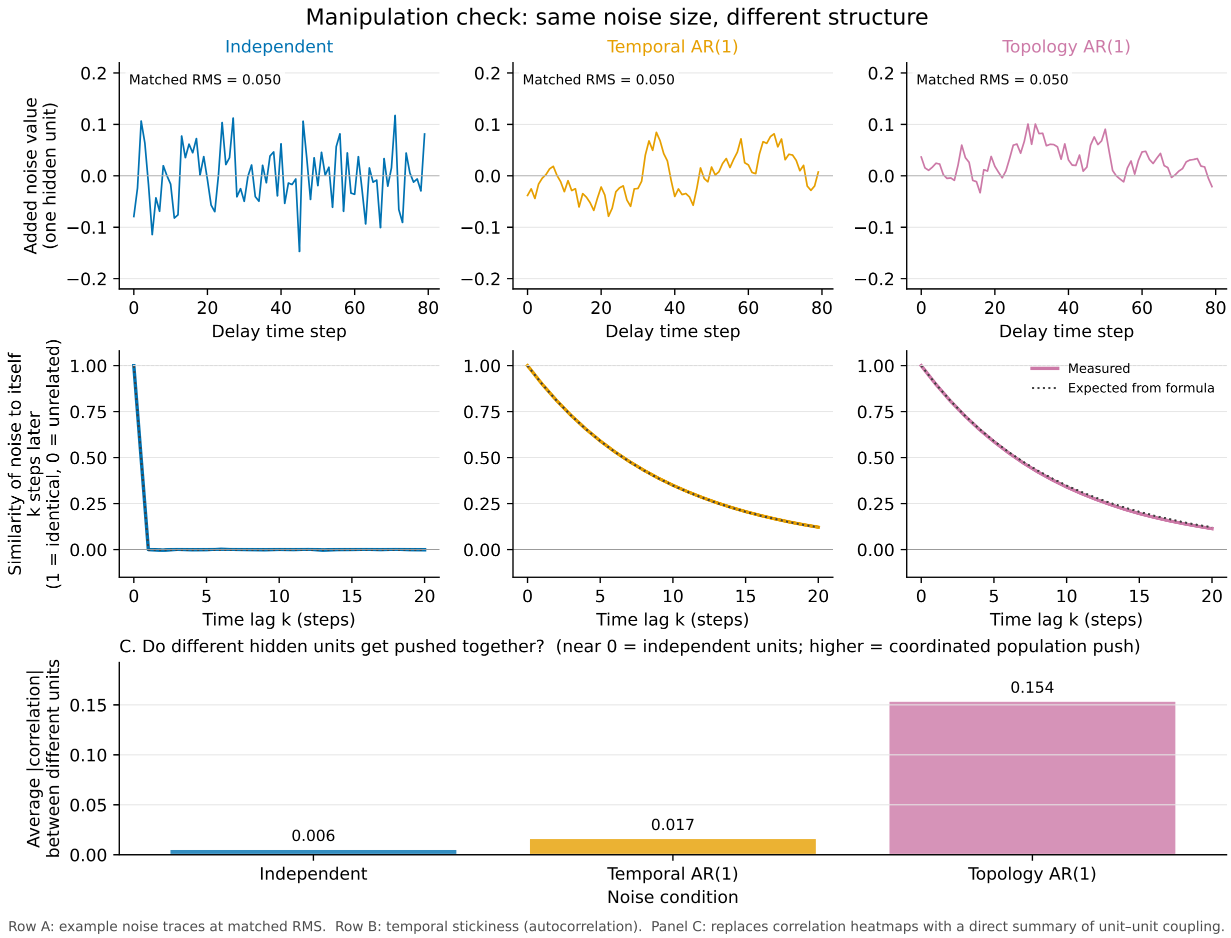


FIG. 2. **Manipulation checks at RMS 0.05.** (A) Example noise on one unit: vertical scale is the added perturbation value; printed RMS confirms amplitude matching. (B) Autocorrelation versus lag: 1 means the noise is identical to itself $k$ steps later; 0 means unrelated. Independent noise falls to $\sim 0$; AR(1) conditions follow $0.9^k$. (C) Average absolute correlation between different hidden units: near zero for independent/temporal; elevated for topology.

(temporal), and $32.43 \pm 4.75°$ (topology). Dose–response at delay 80 preserves the same ordering (Fig. 3B). Fixation gating remained high (Fig. 3C): the model still mostly knew when to wait versus answer; what failed was the remembered angle.

## IV. MAINTENANCE FAILURE AND GEOMETRY

A decoder trained only on clean delay states loses the angle progressively during the hold under structured noise (Fig. 4). End-of-delay geometry explains why: same-angle clouds

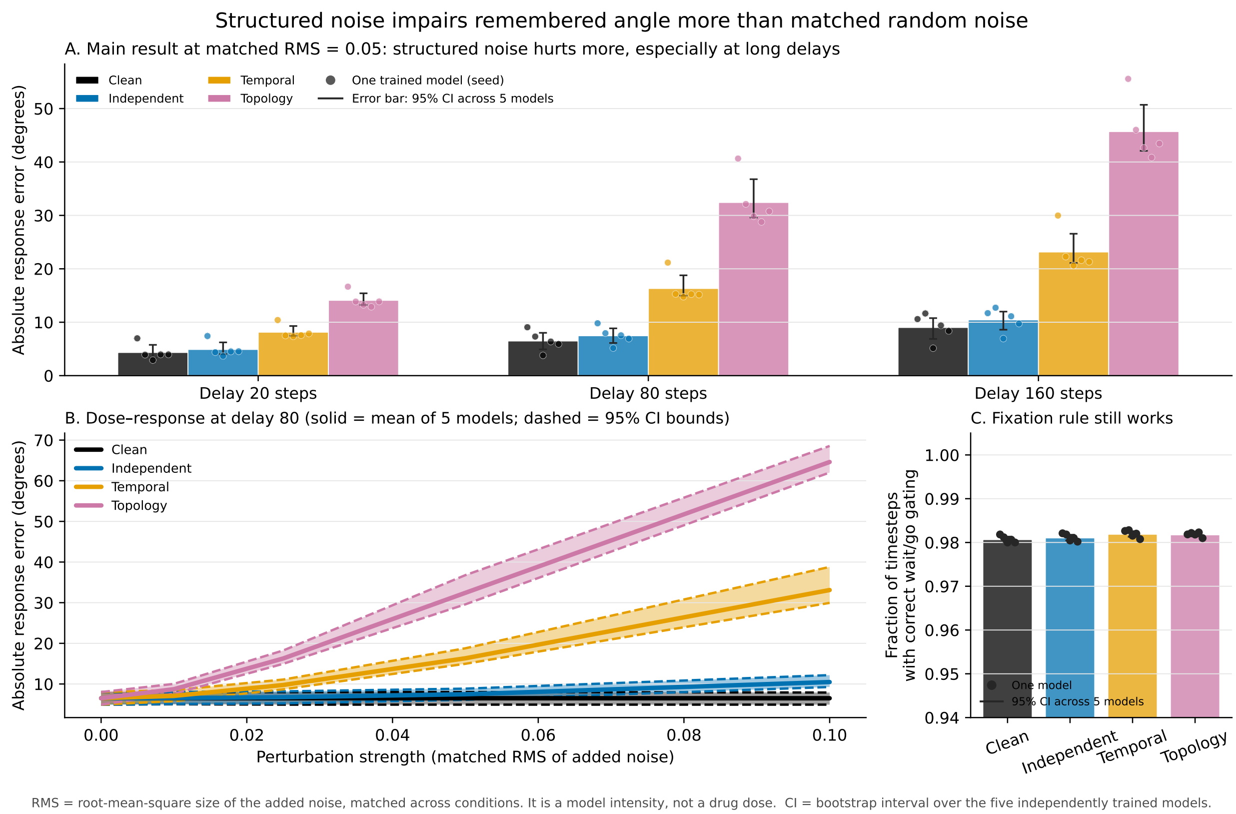


FIG. 3. **Hero comparison.** (A) Response error at RMS 0.05. Coloured points: five independently trained models; bars: five-seed mean; whiskers: 95% bootstrap CI across those five models. (B) Dose–response at delay 80. Solid curves: five-seed means; dashed curves and shaded bands: 95% CI bounds (no individual-seed spaghetti). (C) Fixation accuracy at the same primary point: bars = means, black points = models, whiskers = 95% CI.

TABLE I. Response angular error (°) at RMS 0.05. Mean ± SD across five seed-level means.

| Delay | Clean | Independent | Temporal | Topology |
|---|---|---|---|---|
| 20 | $4.34 \pm 1.54$ | $4.91 \pm 1.45$ | $8.14 \pm 1.28$ | $14.11 \pm 1.48$ |
| 80 | $6.49 \pm 1.93$ | $7.47 \pm 1.68$ | $16.33 \pm 2.71$ | $32.43 \pm 4.75$ |
| 160 | $9.03 \pm 2.50$ | $10.44 \pm 2.24$ | $23.16 \pm 3.86$ | $45.70 \pm 5.82$ |

spread out and different-angle centroids move closer together, most under topology noise (Fig. 5).

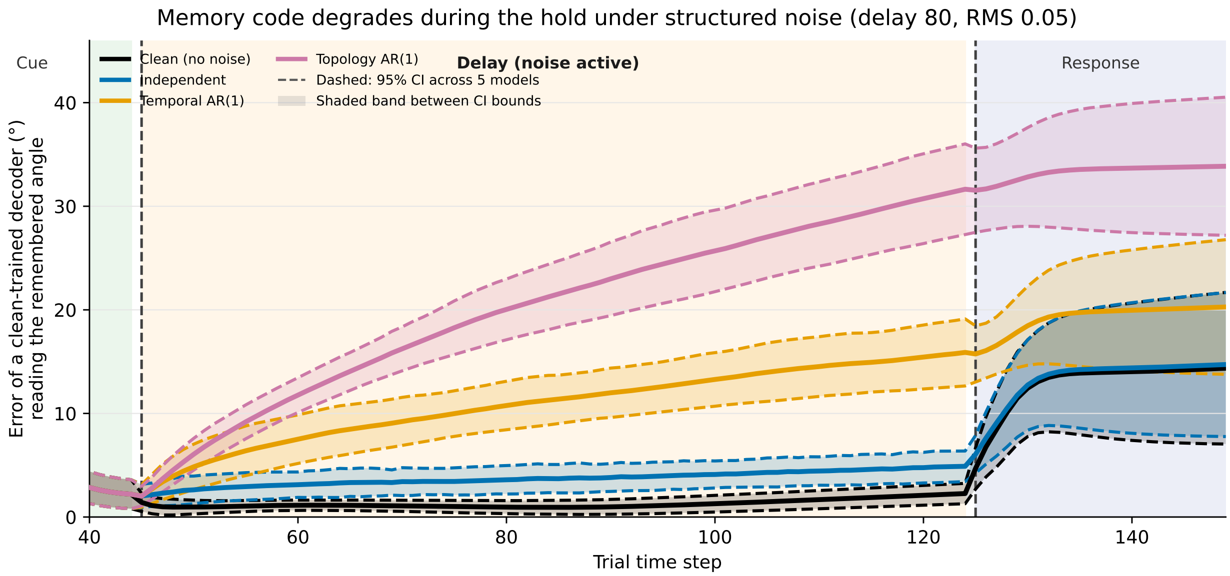


FIG. 4. **Time-resolved decoding at delay 80, RMS 0.05.** Solid: five-seed mean decoder error. Dashed: 95% CI bounds; light fill between bounds. Coloured bands mark cue, delay, and response.

## V. TASK-PHASE SENSITIVITY

Restricting topology-correlated noise to cue, delay, or response shows that maintenance is the vulnerable epoch (Fig. 6). At RMS 0.05 and delay 80, response errors were $7.43 \pm 1.65°$ (cue), $32.12 \pm 4.49°$ (delay), and $11.60 \pm 1.64°$ (response).

## VI. INTERPRETATION BOUNDARY

Equal energy does not imply equal functional impact. In this frozen working-memory RNN, sticky and topology-aligned variability were substantially more damaging than matched independent noise. That constrains candidate psilocybin-informed perturbation designs for this model class; it is not evidence that psilocybin itself is representation-destroying cortical noise.

### DATA AND CODE

Metrics: `outputs/.../noise_structure/full/` and `epoch_timing_full/`.

Figures: `figures/` and `figure_data/`.

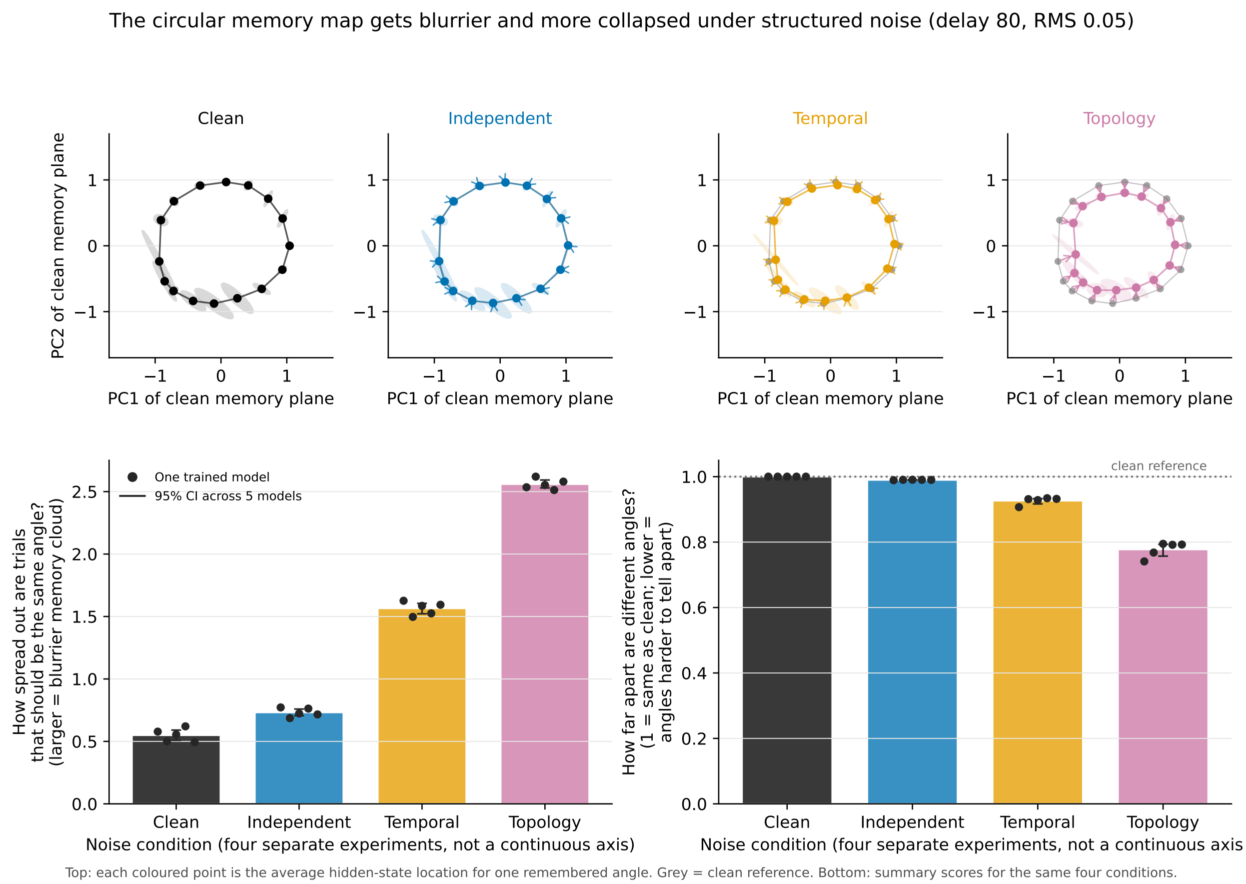


FIG. 5. **Circular memory geometry at delay 80, RMS 0.05.** Top: each panel is one noise condition in the clean-fitted PC1–PC2 plane; coloured points are angle centroids, grey is the clean reference. Bottom: two summary scores across the same four *categorical* conditions (not a continuous left–right axis). Left: within-angle spread. Right: between-angle spacing relative to clean (1 = preserved).

Regenerate: `python-mwm_rnn.noise_structure_dissertation_figures`.

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[1] Z. P. Kilpatrick, B. Ermentrout, and B. Doiron, "Optimizing working memory with heterogeneity of recurrent cortical excitation," J. Neurosci. **33**, 18999–19011 (2013), doi:10.1523/JNEUROSCI.1641-13.2013.

[2] E. Ghazizadeh and S. Ching, "Slow manifolds within network dynamics encode working memory efficiently and robustly," PLoS Comput. Biol. **17**, e1009366 (2021), doi:10.1371/journal.pcbi.1009366.

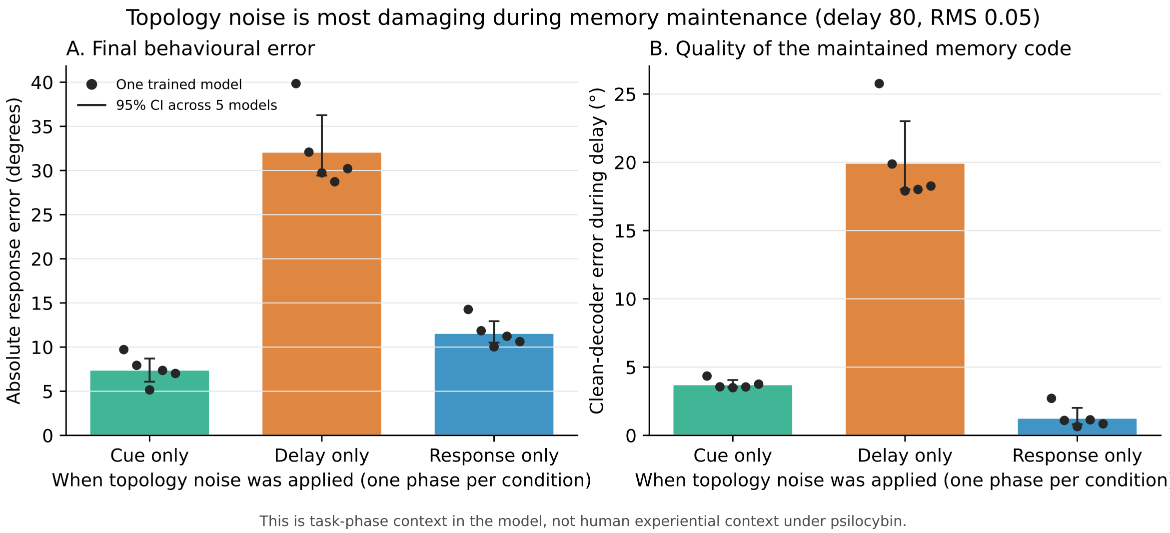


FIG. 6. **Epoch sensitivity for topology AR(1) at delay 80, RMS 0.05.** Bars: five-seed means; black points: individual models; whiskers: 95% CI. Delay-only noise dominates.


[3] R. L. Carhart-Harris *et al.*, "The entropic brain: a theory of conscious states informed by neuroimaging research with psychedelic drugs," Front. Hum. Neurosci. **8**, 20 (2014), doi:10.3389/fnhum.2014.00020.

[4] M. M. Schartner, R. L. Carhart-Harris, A. B. Barrett, A. K. Seth, and S. D. Muthukumaraswamy, "Increased spontaneous MEG signal diversity for psychoactive doses of ketamine, LSD and psilocybin," Sci. Rep. **7**, 46421 (2017), doi:10.1038/srep46421.

[5] R. Herzog *et al.*, "A whole-brain model of the neural entropy increase elicited by psychedelic drugs," Sci. Rep. **13**, 6244 (2023), doi:10.1038/s41598-023-32649-7.

[6] D. Stoliker *et al.*, "Psychedelics align brain activity with context," bioRxiv 2025.03.09.642197, version posted 5 May 2026, doi:10.1101/2025.03.09.642197 (preprint).

[7] C. Bredenberg, F. Normandin, B. Richards, and G. Lajoie, "Modeling the hallucinatory effects of classical psychedelics in terms of replay-dependent plasticity mechanisms," eLife **14**, RP105968 (2026), doi:10.7554/eLife.105968.3.